

FretGarden Pedagogical Constitution

PEDAGOGICAL STANDARD

Chapter 9 - Remediation and Adaptive Difficulty	Version 1.0
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This document forms part of the FretGarden Pedagogical Constitution. Normative terms are used according to the Foundation.

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Chapter 9 - Remediation and Adaptive Difficulty

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1. Purpose

This chapter defines how FretGarden responds when ordinary instruction, practice, feedback, or review does not produce sufficiently stable learning. It establishes the requirements for remediation and for adapting task difficulty without weakening the intended capability.

Remediation is targeted educational action. It is not punishment, repetition without diagnosis, or a permanent alternate curriculum. Adaptive difficulty is the deliberate adjustment of task conditions in response to evidence. It is not the automatic production of easier or harder activities merely to sustain activity.

The purpose of both systems is to restore productive learning while preserving honesty about the learner's current capability. A learner who struggles should receive a better path to the same meaningful objective whenever that objective remains appropriate. A learner should not be advanced through hidden simplification, trapped in endless repetition, or forced to repeat material that current evidence already supports.

This chapter establishes:

- 1 conditions that trigger remediation;
- 2 distinctions among lapse, misconception, prerequisite failure, execution breakdown, overload, and task-design failure;
- 3 the remediation cycle;
- 4 rules for selecting the smallest effective intervention;
- 5 adaptive difficulty variables and their permissible use;
- 6 criteria for reducing, maintaining, or increasing challenge;
- 7 rules for preserving objective integrity;
- 8 return-to-path and revalidation requirements;
- 9 learner agency, transparency, and dignity;
- 10 and constitutional conformance standards.

2. Constitutional Authority

This chapter operationalizes the following commitments:

- 1 prerequisite failure **MUST** be remediable;
- 2 repeated error patterns **MUST** change the instructional response;

- 3 difficulty MUST serve learning rather than appearance of rigor;
- 4 support MUST be temporary and faded deliberately;
- 5 progression MUST depend on readiness rather than completion;
- 6 review and reactivation MUST be distinguished from first-time instruction;
- 7 feedback MUST lead to an actionable next attempt;
- 8 error classifications MUST not exceed available evidence;
- 9 learner-facing status language MUST remain honest;
- 10 and adaptation MUST develop independence rather than permanent dependence.

Where another specification treats remediation as repetition, punishment, content demotion, or score recovery alone, this chapter governs.

3. Governing Principle

Remediation MUST address the most defensible cause of unstable performance with the least disruptive intervention capable of restoring productive learning.

Adaptive difficulty MUST change the conditions of learning without silently changing the capability being claimed.

The governing sequence is:

- 1 identify the objective and required evidence;
- 2 verify that the task and instructions were valid;
- 3 determine whether evidence is missing, negative, contradictory, or stale;
- 4 classify the most likely barrier with appropriate uncertainty;
- 5 identify the smallest relevant variable that can be changed;
- 6 provide targeted instruction, practice, support, or review;
- 7 require a corrected or reconstructed attempt;
- 8 fade temporary support;
- 9 re-test the original or equivalent objective conditions;
- 10 and return the learner to ordinary progression only when readiness evidence is sufficient.

4. Core Distinctions

4.1 Remediation Is Not Repetition

Repeating the same unsuccessful task without changing explanation, support, scope, tempo, representation, or strategy is not remediation.

4.2 Easier Is Not Always More Appropriate

A task may be difficult because of a missing prerequisite, unclear instructions, excessive cognitive load, physical instability, or an invalid assessment. Lowering the overall challenge may conceal rather than resolve the cause.

4.3 Harder Is Not Always More Advanced

Adding speed, choices, strings, keys, distractions, or musical context does not necessarily produce deeper learning. Difficulty must correspond to a defined transfer or fluency demand.

4.4 Lapse Is Not Initial Absence

A previously demonstrated capability that has weakened should be reactivated differently from a capability that was never established.

4.5 Support Is Not Mastery

Success under increased support demonstrates supported recovery. Independent evidence is still required before a strong capability claim is restored.

4.6 Adaptation Is Not Concealment

FretGarden MUST not make a task easier while continuing to describe the learner as having met the original standard.

4.7 Learner Struggle Is Not Automatically Learner Deficit

The activity, explanation, representation, pacing, instrument conditions, or measurement method may be responsible. Task validity must be examined before the learner is classified.

5. Definitions

5.1 Remediation

Remediation is targeted instruction or practice intended to repair an unstable prerequisite, misconception, error pattern, lapsed capability, or strategy failure.

5.2 Adaptive Difficulty

Adaptive difficulty is the evidence-based adjustment of task conditions to preserve productive challenge and improve the validity of learning.

5.3 Barrier

A barrier is a condition that prevents the learner from making a meaningful attempt or demonstrating the intended capability.

5.4 Bottleneck

A bottleneck is the component capability that most constrains an integrated task.

5.5 Prerequisite Failure

Prerequisite failure occurs when a required earlier capability is insufficient for the present task.

5.6 Misconception

A misconception is an internally coherent but musically inaccurate model that produces patterned errors.

5.7 Lapse

A lapse is reduced access to a capability that prior evidence showed was once available.

5.8 Reactivation

Reactivation is the restoration of a lapsed capability through retrieval, cueing, reconstruction, or concise review.

5.9 Scaffold

A scaffold is temporary support that enables productive action beyond the learner's current independent performance.

5.10 Difficulty Variable

A difficulty variable is a controllable condition that changes task demand without necessarily changing the target objective.

5.11 Objective Integrity

Objective integrity is the preservation of the capability, content scope, and evidence claim while task conditions are adapted.

5.12 Return-to-Path

Return-to-path is the transition from remediation back to ordinary lesson, review, or progression flow after sufficient evidence has been restored.

6. Purposes of Remediation

Remediation MAY serve one or more of the following purposes:

- 1 restore a missing prerequisite;
- 2 correct a misconception;
- 3 stabilize retrieval;
- 4 reduce physical or temporal complexity;
- 5 repair an ineffective practice strategy;
- 6 re-establish attention to the target feature;
- 7 reactivate lapsed knowledge;
- 8 distinguish learner difficulty from task-design failure;
- 9 calibrate an appropriate challenge level;
- 10 and prepare the learner to reattempt the original objective.

Remediation MUST have an exit condition. It MUST NOT become an indefinite holding state.

7. Remediation Triggers

A remediation review SHOULD be triggered when one or more of the following occur:

- 1 the same error pattern persists after ordinary feedback and reattempt;
- 2 performance remains unstable after scaffold fading;
- 3 a required prerequisite fails activation;

- 4 delayed review shows substantial loss of access;
- 5 near-transfer tasks fail despite success on identical items;
- 6 the learner repeatedly guesses rather than retrieves;
- 7 tempo, movement, or cognitive demand causes breakdown below the required conditions;
- 8 self-evaluation and observed performance remain substantially miscalibrated;
- 9 the learner cannot explain or reconstruct a relationship central to later use;
- 10 or evidence becomes contradictory enough that progression is unsafe.

A single error SHOULD NOT automatically trigger extensive remediation unless the error reveals a dangerous misconception, a critical prerequisite absence, or a task-design problem that prevents valid continuation.

8. Before Remediating the Learner

FretGarden MUST first evaluate whether the task itself is responsible.

Reviewers should ask:

- 1 Were the instructions unambiguous?
- 2 Did the task measure the stated capability?
- 3 Was required information available?
- 4 Did the representation introduce an unrelated demand?
- 5 Was the tempo or response window appropriate?
- 6 Was the instrument assumed to be in a valid condition?
- 7 Could chance, interface behavior, or visual patterning distort the result?
- 8 Was the learner given a meaningful opportunity to respond?
- 9 Did the feedback identify the correct dimension?
- 10 and was the prerequisite expectation itself justified?

When task invalidity is plausible, the task MUST be corrected or replaced before the learner is assigned remediation for that evidence.

9. Barrier Classification

Remediation decisions SHOULD distinguish among the following barrier families.

9.1 Orientation Barrier

The learner does not understand the representation, terminology, directions, or task procedure.

9.2 Perceptual Barrier

The learner cannot yet detect or distinguish the relevant sound or temporal feature.

9.3 Retrieval Barrier

The knowledge is not accessible with the required degree of support or speed.

9.4 Conceptual Barrier

The learner lacks an accurate model or applies an incorrect rule.

9.5 Navigation Barrier

The learner cannot locate or connect the relevant relationship on the fretboard.

9.6 Execution Barrier

The learner understands the target but cannot perform the movement reliably.

9.7 Temporal Barrier

The learner cannot maintain pulse, subdivision, placement, or synchronization.

9.8 Integration Barrier

Component skills are present separately but break down when combined.

9.9 Strategy Barrier

The learner uses an ineffective practice or problem-solving method.

9.10 Attention Barrier

Evidence suggests the capability is available but inconsistently applied because attention has degraded or the task has become careless.

9.11 Retention Barrier

A capability was previously available but is no longer reliably retrievable after delay.

9.12 Accessibility or Condition Barrier

The standard response mode, physical demand, sensory demand, or environment prevents a valid demonstration.

Barrier classifications MUST remain hypotheses when evidence is incomplete.

10. The Remediation Contract

Every formal remediation path MUST define:

- 1 the original objective;
- 2 the evidence that triggered remediation;
- 3 the suspected barrier and confidence level;
- 4 the prerequisite or component capability being targeted;
- 5 the intervention selected;
- 6 the difficulty variables changed;
- 7 the support introduced;
- 8 the success criterion within remediation;
- 9 the scaffold-fading plan;
- 10 the revalidation task;

- 11 the return-to-path rule;
- 12 and the conditions that require escalation or redesign.

11. The Remediation Cycle

A compliant remediation cycle consists of:

- 1 Verify - confirm the objective, task validity, and evidence conditions.
- 2 Localize - identify the smallest plausible bottleneck.
- 3 Reduce - remove unrelated demands while preserving the target relationship.
- 4 Re-explain or Reconstruct - provide the minimum instruction needed to repair the model or procedure.
- 5 Practice Correctly - create controlled successful attempts.
- 6 Vary - confirm that success is not tied to one item.
- 7 Fade - remove temporary support.
- 8 Revalidate - assess the original or equivalent conditions.
- 9 Return or Escalate - resume ordinary progression or select a deeper intervention.

Skipping revalidation is prohibited when remediation affects a prerequisite or mastery claim.

12. Minimum Effective Intervention

FretGarden SHOULD use the least intensive intervention likely to resolve the barrier.

A typical escalation order is:

- 1 restate the target;
- 2 direct attention to the relevant dimension;
- 3 provide a limited cue;
- 4 reduce one difficulty variable;
- 5 isolate the failing component;
- 6 provide a worked comparison or model;
- 7 reconstruct the prerequisite;
- 8 change representation;
- 9 provide a dedicated remedial sequence;
- 10 or route to a broader prerequisite review.

The escalation order MAY vary by domain. Complete answer revelation should not be the default first response.

13. Preserving Objective Integrity

Adaptation may change conditions while preserving the target capability.

For example, a learner practicing chord-tone location may receive:

- a smaller fret range;
- fewer chord types;
- slower harmonic rhythm;
- one visible root anchor;
- or reduced simultaneous choices.

The objective has been weakened if the adaptation removes the essential decision, relationship, or evidence demand.

An adaptation **MUST** be classified as one of the following:

- 1 Condition-preserving adaptation - the objective and claim remain unchanged.
- 2 Scaffolded adaptation - the objective remains, but evidence is supported and cannot establish independence.
- 3 Scope-reduced adaptation - a narrower objective is temporarily practiced and must later be expanded.
- 4 Objective substitution - a different prerequisite objective is taught. This must be explicit.

14. Difficulty Variables

Difficulty **MAY** be adjusted through the following variables when relevant.

14.1 Content Scope

Number of notes, intervals, chord types, strings, positions, keys, rhythms, or concepts included.

14.2 Response Set

Number and similarity of possible answers.

14.3 Prompting

Amount of visual, verbal, auditory, or procedural support.

14.4 Time Demand

Tempo, response window, harmonic rhythm, or duration of sustained attention.

14.5 Physical Demand

Distance, string crossing, position shifting, coordination, articulation, or endurance.

14.6 Cognitive Integration

Number of capabilities used simultaneously.

14.7 Representational Demand

Need to translate among sound, fretboard, notation, tablature, interval formula, scale degree, and language.

14.8 Context Familiarity

Exact practiced item, near variation, unfamiliar example, or novel integration.

14.9 Interference

Presence of similar alternatives, mixed categories, distractors, or interleaved material.

14.10 External Pulse and Support

Density of metronome clicks, backing context, count-in, reference pitch, or visual anchor.

Difficulty variables **MUST** be changed deliberately. During diagnosis, changing several variables at once **SHOULD** be avoided because the cause of improvement becomes unclear.

15. Productive Difficulty States

15.1 Under-Challenged

The learner succeeds consistently with little attention and no meaningful variation. The task may be ready for reduced support, greater variation, or application.

15.2 Productively Challenged

The learner makes meaningful attempts, errors are interpretable, correction improves performance, and success remains attainable.

15.3 Unstable but Recoverable

The learner experiences repeated breakdown but can improve through targeted support. Local remediation is appropriate.

15.4 Overloaded

Errors occur across several dimensions, attempts are difficult to interpret, or the learner cannot apply feedback. Demand should be decomposed or reduced.

15.5 Blocked by Prerequisite

The task cannot proceed meaningfully until an earlier capability is restored.

15.6 Invalidly Challenged

Difficulty comes primarily from unclear representation, unrelated demands, inaccessible response mode, or faulty task design. The task must be corrected.

16. Rules for Reducing Difficulty

Difficulty **SHOULD** be reduced when:

- 1 the learner cannot produce interpretable attempts;
- 2 repeated errors continue after appropriate feedback;
- 3 the target is obscured by simultaneous novel demands;
- 4 tempo causes loss of accuracy or control;
- 5 support was faded before stability existed;
- 6 fatigue or attention deterioration makes practice careless;
- 7 or a narrower scope is required to expose the barrier.

Reduction **MUST** preserve a clear route back to the original conditions.

17. Rules for Increasing Difficulty

Difficulty **SHOULD** increase only when current evidence shows stable success under the present conditions and the change serves a defined goal such as:

- cue fading;
- expanded scope;
- near transfer;
- interleaving;
- tempo development;
- contextual application;
- or integrated musicianship.

Difficulty **MUST NOT** increase solely because a fixed number of attempts has occurred.

18. Maintaining Difficulty

Maintaining the current challenge is appropriate when the learner is productively challenged and evidence is still developing.

FretGarden **SHOULD** avoid unnecessary oscillation between easier and harder tasks. Adaptation should be responsive without becoming erratic.

19. Tempo Adaptation

Tempo is one difficulty variable, not the universal remedy.

Tempo should be reduced when the learner cannot preserve required note accuracy, rhythmic placement, tone, control, or low unnecessary tension.

Tempo should increase only when:

- 1 the current tempo is stable across repeated attempts;
- 2 the learner can recover from minor disruption;
- 3 the target movement or rhythm remains clear;
- 4 and the increase is small enough to preserve control.

A tempo reduction does not weaken the objective when tempo is not yet part of the mastery claim. When tempo is part of the objective, slower performance is supported evidence, not completion of the original standard.

20. Scope Reduction

Scope reduction isolates a manageable subset of the original objective.

Examples include:

- two strings instead of six;

- one interval class before mixed intervals;
- one measure before a full phrase;
- one chord transition before a progression;
- or one scale-degree function before a complete set.

Scope reduction **MUST** identify what remains untested and must include a later expansion plan.

21. Representation Change

A different representation may reveal or remove a barrier.

For example:

- sound before notation;
- interval formula before fretboard shape;
- clapping before instrumental execution;
- verbal explanation before speeded recall;
- or a physical demonstration before independent movement.

Representation change must clarify the same relationship or explicitly teach a prerequisite. It must not create the appearance of success by replacing the intended construct.

22. Scaffold Introduction and Fading

Scaffolds may include:

- highlighted roots;
- reduced answer sets;
- verbal cues;
- partial diagrams;
- count-ins;
- slower pulse;
- worked first examples;
- isolated components;
- or guided self-questioning.

Every scaffold **MUST** have:

- 1 a purpose;
- 2 a record of what it reveals or conceals;
- 3 a fading condition;
- 4 and an independent reattempt after removal.

A scaffold that remains indefinitely should be reconsidered as part of the task definition rather than mislabeled as temporary support.

23. Misconception Remediation

Misconceptions require conceptual repair, not merely more attempts.

A compliant sequence SHOULD:

- 1 surface the learner's current model;
- 2 identify the specific contradiction or boundary;
- 3 provide a clear counterexample or contrast;
- 4 reconstruct the accurate relationship;
- 5 apply it to several examples;
- 6 require explanation or prediction;
- 7 and verify that the old error pattern no longer controls performance.

Directly supplying isolated correct answers is insufficient.

24. Retrieval Remediation

Retrieval weakness may require:

- reduced cueing rather than complete re-teaching;
- reconstruction from known anchors;
- smaller item sets;
- increased spacing;
- comparison of confusable items;
- or more frequent low-stakes retrieval.

When the learner can derive an answer but not retrieve it fluently, the capability should be classified as developing rather than absent.

25. Perceptual Remediation

Perceptual remediation SHOULD begin with clear contrasts and controlled contexts.

It may adjust:

- interval size contrast;
- register;
- timbre;
- harmonic context;
- tempo;
- repetition;
- response set;
- or the delay between examples.

Visual information must not allow the learner to bypass the auditory target.

26. Fretboard-Navigation Remediation

Navigation remediation may use:

- stable anchor notes;
- smaller fret regions;
- one string set;
- relative interval paths;
- reconstruction from roots;
- connection between adjacent positions;
- and gradual removal of highlighted locations.

The intervention **MUST** avoid converting a relational objective into permanent diagram copying.

27. Physical-Execution Remediation

Execution remediation **SHOULD** isolate the smallest movement or transition responsible for breakdown.

It may adjust:

- tempo;
- movement range;
- fingering choice;
- number of repetitions;
- articulation;
- coordination layers;
- or phrase length.

Feedback must remain within pedagogical observation and **MUST NOT** diagnose injury. Pain or persistent physical distress requires stopping and appropriate professional guidance.

28. Temporal Remediation

Temporal remediation may separate:

- pulse;
- subdivision;
- counting;
- rhythmic pattern;
- note selection;
- and instrumental movement.

A learner may clap or count before playing. Metronome density may be increased temporarily and then reduced. The final evidence must return to the required musical conditions.

29. Integration Remediation

When component capabilities are present separately but fail in combination, remediation SHOULD:

- 1 identify the first point of breakdown;
- 2 reduce the number of simultaneous demands;
- 3 recombine components gradually;
- 4 preserve the musical relationship;
- 5 and verify the complete task after each component is restored.

Repeatedly practicing the full failing task is not sufficient.

30. Metacognitive Remediation

When the learner uses ineffective practice strategies, FretGarden should teach the decision process directly.

The learner may be required to:

- identify the target;
- choose a manageable tempo;
- isolate the error;
- select a correction strategy;
- define a successful repetition;
- and determine when to stop or increase difficulty.

The goal is not only corrected performance but improved future self-regulation.

31. Lapse and Reactivation

A lapsed capability should first receive a retrieval opportunity.

If retrieval fails, support should escalate from cues to reconstruction to concise re-teaching. Previously mastered content SHOULD NOT automatically restart from the most elementary instruction.

Reactivation evidence must be distinguished from new mastery evidence. Critical capabilities may require delayed confirmation after reactivation.

32. Revalidation

Revalidation confirms whether remediation restored the intended capability.

A valid revalidation task MUST:

- 1 assess the original objective or an equivalent form;
- 2 remove temporary supports not permitted by the claim;
- 3 use representative rather than identical items where transfer matters;
- 4 preserve required quality dimensions;

- 5 and record whether success was immediate, corrected, or independent.

Remediation is not complete until the learner either passes revalidation, enters a revised scope with explicit status, or is routed to a deeper intervention.

33. Return-to-Path Rules

The learner may return to ordinary progression when:

- 1 the identified barrier has been sufficiently repaired;
- 2 required prerequisites meet the readiness threshold;
- 3 scaffolded success has been followed by independent evidence;
- 4 the original task conditions are again productive;
- 5 and the next lesson does not assume capability beyond the restored scope.

Return-to-path SHOULD be quiet and nonpunitive. The learner need not be told they have escaped a failure state.

34. Escalation Rules

Escalation is required when:

- the same barrier persists through a complete local cycle;
- several prerequisites are unstable;
- evidence suggests a broad misconception;
- the response mode is inaccessible;
- the task remains invalid;
- or repeated adaptation produces only scaffold-dependent success.

Escalation may include a dedicated prerequisite lesson, alternate representation, broader diagnostic sequence, human review, accessibility accommodation, or curriculum-path reconsideration.

35. Avoiding Remediation Loops

A learner MUST NOT be trapped in indefinite repetition.

A remediation path SHOULD include limits on:

- identical retries;
- repeated hint cycles;
- time spent without improvement;
- and number of escalations before redesign or alternate routing.

When the intervention does not improve evidence, the system must change its hypothesis or task.

36. Adaptive Sequencing

Adaptive sequencing may alter which objective appears next based on readiness and evidence.

It MAY:

- insert prerequisite review;
- delay a dependent lesson;
- select a parallel application;
- provide additional consolidation;
- or skip redundant instruction after valid diagnostic evidence.

Adaptive sequencing **MUST** preserve curriculum coherence and make prerequisite assumptions explicit.

37. Learner Choice and Agency

Learners **SHOULD** be allowed meaningful choices when several interventions are pedagogically valid.

Possible choices include:

- slower tempo or smaller scope;
- visual reconstruction or verbal explanation;
- immediate retry or short break;
- guided review or diagnostic check;
- and one of several musically equivalent examples.

Choice must not require the learner to diagnose a problem they are not yet equipped to interpret.

38. Transparency

FretGarden should explain adaptations in direct, nonjudgmental language.

Examples include:

- "This interval is stable in one position but not yet across positions. We will narrow the fretboard range and rebuild the relationship.";
- "The notes are correct, but the subdivision is unstable. We will keep the notes and simplify the rhythm.";
- "This looks like a recall lapse rather than a new concept. Try reconstructing it from the root before reviewing the answer.";
- and "The current task is not giving clear evidence, so we will check the prerequisite separately."

The system **SHOULD** avoid opaque changes that make the learner feel arbitrarily rewarded or blocked.

39. Dignity and Emotional Safety

Remediation **MUST NOT** be framed as punishment, demotion, failure of talent, or loss of worth.

Language such as "basic," "behind," or "failed" should be used only when technically necessary and carefully contextualized. Preferred language describes the capability and next action.

The learner should understand that revisiting foundations is normal maintenance within the Garden Philosophy.

40. Evidence Records

A remediation record SHOULD preserve:

- objective;
- trigger evidence;
- suspected barrier;
- intervention;
- support level;
- difficulty variables changed;
- performance during intervention;
- revalidation result;
- remaining uncertainty;
- and next review obligation.

The exact data implementation is outside this chapter. The pedagogical requirement is that supported success not erase the conditions under which it occurred.

41. Mastery and Progression Effects

Remediation MAY alter progression status, but it MUST NOT erase valid prior evidence without cause.

A learner may remain:

- ready with support;
- ready within a reduced scope;
- consolidating;
- temporarily not ready for a dependent objective;
- or revalidated after lapse.

Progression decisions must follow Chapter 6 and review obligations must follow Chapter 7.

42. Domain-Specific Adaptation

Later domain chapters MUST identify:

- 1 which difficulty variables are valid;
- 2 which adaptations preserve the objective;
- 3 common barrier patterns;
- 4 appropriate intervention sequences;
- 5 revalidation requirements;
- 6 and prohibited shortcuts.

No domain chapter may weaken the requirement that the original capability be re-demonstrated after temporary support.

43. Anti-Patterns

The following practices are noncompliant:

- 1 repeating the identical failed item indefinitely;
- 2 lowering difficulty without recording the changed claim;
- 3 increasing difficulty after an arbitrary repetition count;
- 4 treating all errors as prerequisite failure;
- 5 revealing answers before allowing reconstruction;
- 6 using easier distractors to manufacture success;
- 7 routing every lapse through complete re-teaching;
- 8 preserving permanent visual anchors while claiming independent navigation;
- 9 changing several diagnostic variables at once without reason;
- 10 adapting only to accuracy while ignoring rhythm, control, or understanding;
- 11 using remediation as punishment for low engagement;
- 12 and allowing adaptive systems to override constitutional progression standards.

44. Constitutional Non-Negotiables

- 1 Remediation **MUST** be triggered by evidence, not assumption alone.
- 2 Task validity **MUST** be considered before learner deficit is inferred.
- 3 Every remediation path **MUST** identify the original objective.
- 4 Repetition without changed instructional conditions **MUST NOT** be labeled remediation.
- 5 Adaptive difficulty **MUST** identify which variables changed.
- 6 Supported performance **MUST NOT** be reported as independent mastery.
- 7 Scope reduction **MUST** identify what remains outside the claim.
- 8 Temporary scaffolds **MUST** include fading and independent reattempt.
- 9 Misconceptions **MUST** receive conceptual repair, not answer repetition alone.
- 10 Critical prerequisite repair **MUST** include revalidation.
- 11 Difficulty increases **MUST** serve a defined learning goal.
- 12 Learners **MUST NOT** be trapped in indefinite retry loops.
- 13 Adaptation **MUST** preserve learner dignity and honest status.
- 14 Accessibility barriers **MUST NOT** be misclassified as lack of musical capability.
- 15 Return-to-path **MUST** depend on evidence appropriate to future demands.
- 16 Remediation records **SHOULD** preserve support, scope, and remaining uncertainty.
- 17 Learner choice **SHOULD** be provided when several valid interventions exist.
- 18 Adaptive systems **MUST** remain subordinate to the constitution.

45. Conformance Tests

A remediation or adaptive design conforms only if reviewers can answer:

- 1 What objective is unstable?
- 2 What evidence triggered intervention?
- 3 Was the task itself validated?
- 4 What barrier is suspected, and with what confidence?
- 5 What is the smallest relevant bottleneck?
- 6 Which variable is being changed?
- 7 Does the change preserve the objective?
- 8 What support is introduced?
- 9 How will support be faded?
- 10 What counts as improvement during remediation?
- 11 How will the original capability be revalidated?
- 12 What happens if the intervention fails?
- 13 How is the learner protected from endless repetition?
- 14 What claim is permitted during supported success?
- 15 When may ordinary progression resume?
- 16 How will the capability be reviewed after return?
- 17 Does the intervention develop future self-correction?
- 18 and is the learner-facing explanation accurate and respectful?

46. Tradeoffs and Resolution Rules

46.1 Personalization Versus Coherence

Adaptation should respond to the learner, but each path must remain aligned with shared objectives and standards. Personalization may vary route, support, pacing, and examples; it must not silently create incompatible definitions of mastery.

46.2 Immediate Success Versus Durable Repair

Answer revelation or drastic simplification may restore immediate success but fail to repair the underlying capability. Durable repair takes precedence when the objective is core.

46.3 Efficiency Versus Diagnostic Confidence

Rapid intervention is useful, but premature diagnosis may send the learner down the wrong path. Use low-cost reversible interventions first when confidence is limited.

46.4 Challenge Versus Sustainability

Productive challenge is required, but excessive struggle degrades attention and practice quality. The target is interpretable effort, not maximal difficulty.

46.5 Automation Versus Human Judgment

Automated adaptation may respond consistently to observable patterns but cannot infer every cause. Uncertain, high-stakes, accessibility-sensitive, or persistently contradictory cases may require human review.

46.6 Progression Momentum Versus Prerequisite Repair

Inserting remediation may slow visible advancement, but proceeding with an unstable required prerequisite creates greater cumulative difficulty. Functional readiness takes precedence over superficial momentum.

47. Assumptions Made

This chapter assumes:

- 1 objectives and evidence claims follow Chapter 2;
- 2 lessons and exercises follow Chapters 3 and 4;
- 3 assessments follow Chapter 5;
- 4 mastery and progression follow Chapter 6;
- 5 review and reactivation follow Chapter 7;
- 6 feedback and error classifications follow Chapter 8;
- 7 some adaptive decisions can be made from limited evidence but must preserve uncertainty;
- 8 domain-specific chapters will refine intervention strategies;
- 9 learners may progress unevenly across objective domains;
- 10 and remediation is expected to be a normal part of long-term learning.

48. Unresolved Questions

The following matters remain unresolved:

- 1 default thresholds for entering local versus extended remediation;
- 2 standard limits for identical retries;
- 3 confidence requirements for automated barrier classification;
- 4 domain-specific adaptation step sizes;
- 5 rules for learner-requested challenge changes;
- 6 handling of persistent disagreement between self-report and observed evidence;
- 7 standards for human instructor override;
- 8 accessibility-specific equivalent pathways;
- 9 adaptive sequencing across optional curriculum branches;
- 10 evidence decay after successful remediation;
- 11 governance of adaptive algorithms and educational analytics;
- 12 and learner-facing visualization of remediation without stigmatization.

49. Topics Intentionally Deferred

Deferred topics include:

- 1 domain-specific remedial libraries;
- 2 adaptive algorithm implementation;
- 3 statistical confidence models;
- 4 educational analytics;
- 5 accessibility accommodations;
- 6 human instructor workflows;
- 7 placement and bypass systems;
- 8 practice recommendation generation;
- 9 learner preference settings;
- 10 detailed motivational interventions;
- 11 curriculum graph implementation;
- 12 and quality-assurance procedures for adaptive content.

50. Required Future Cross-References

50.1 Domain-Specific Pedagogy Chapters

Must define valid barrier subtypes, difficulty variables, remediation sequences, and revalidation standards for their domains.

50.2 Accessibility Chapter

Must define equivalent demonstrations and adaptations that separate access barriers from musical capability.

50.3 Educational Analytics Chapter

Must distinguish raw performance, inferred barrier, confidence, intervention, and outcome.

50.4 Practice Design Chapter

Must reference strategy remediation, session regulation, tempo adjustment, stopping conditions, and learner agency.

50.5 Curriculum Sequencing Chapter

Must define how remediation interacts with prerequisite graphs, optional branches, and return-to-path.

50.6 Constitutional Governance Chapter

Must define review, audit, and amendment procedures for adaptive rules and intervention libraries.

51. Chapter Conformance Statement

A FretGarden remediation system conforms to this chapter only when it responds to defensible evidence, distinguishes learner barriers from task invalidity, selects a targeted intervention, records changed conditions, fades temporary support, revalidates the original capability, and returns the learner to ordinary progression without false claims or stigma.

An adaptive difficulty system conforms only when it changes difficulty for an explicit pedagogical purpose, preserves objective integrity, prevents unproductive overload and trivial success, and remains subordinate to the constitution's standards for evidence, mastery, review, and learner independence.

Remediation is successful not when the learner merely produces a correct answer under easier conditions, but when the relevant capability becomes available again under conditions appropriate to its intended future use.

End of Chapter 9 - Remediation and Adaptive Difficulty